



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
11-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

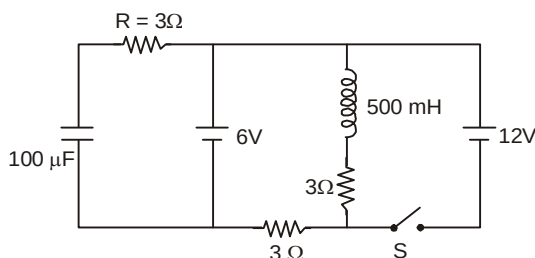
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

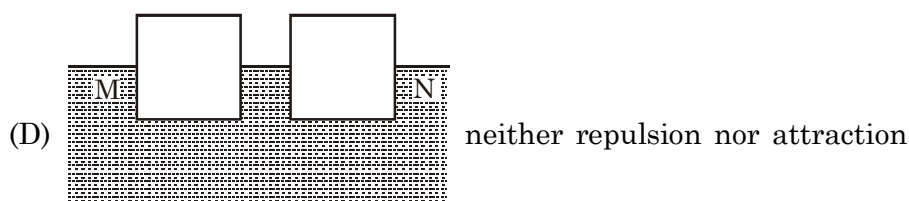
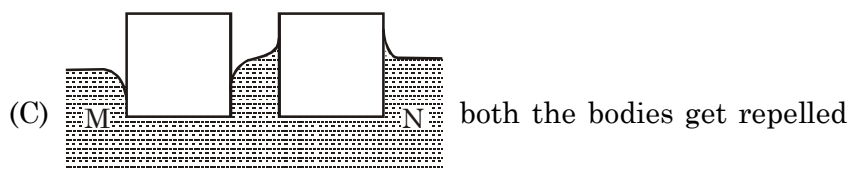
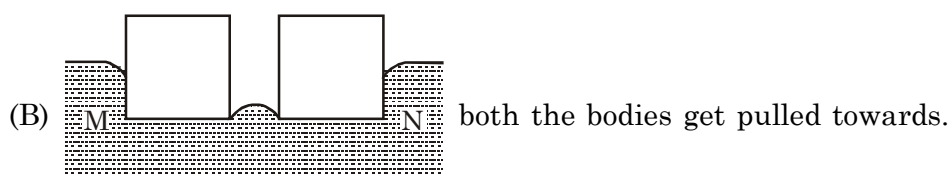
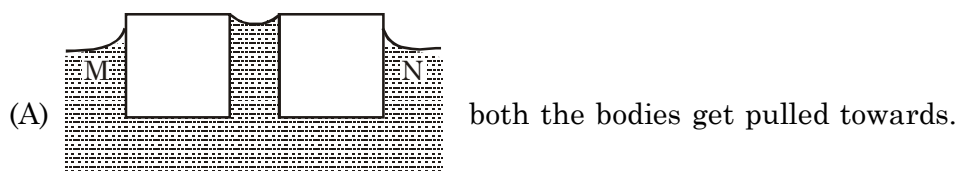
1. In the given circuit switch is closed at $t = 0$,



- (A) Current in the inductor when the circuit reaches the steady state is 4 A.
 (B) The net change in flux in the inductor is 1.5 Wb.
 (C) The time constant of the circuit after closing S is 555.55 s.
 (D) The charge stored in the capacitor in the steady state is 1.2 mC.

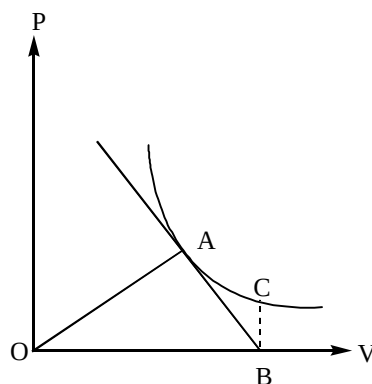
Space for Rough Work

2. In the Bohr model of the atom (Here n is the principal quantum number)
- (A) Frequency of revolution of the electron is inversely proportional to n^3
- (B) The modulus of potential energy of the electron in an orbit is greater than its kinetic energy.
- (C) Magnetic induction at the nucleus produced due to the motion of electron motion in the n^{th} orbit is proportional to n^5 .
- (D) Magnetic moment produced due to the motion of electron motion in the n^{th} orbit is proportional to n .
3. Two light bodies are kept floating on the surface of water very close to each other as shown below. Water surface takes different shapes depending on the surface nature of the bodies. Identify true situations from the following arrangement in the view of surface tension : Assume complete wetting/ $\theta = 90^\circ$ / no wetting as the case may be.



Space for Rough Work

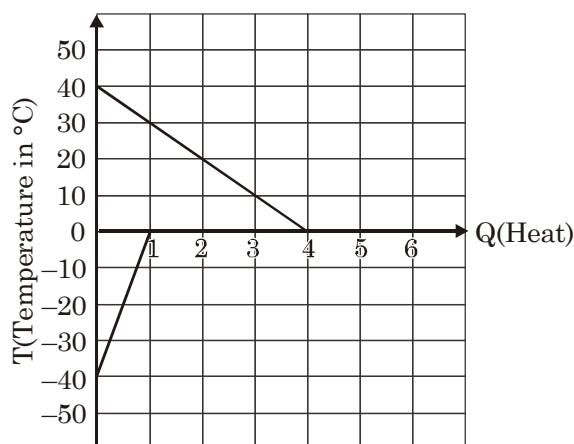
4. n moles of an ideal gas undergo an isothermal process at temperature T . P-V graph of the process is as shown in the figure. A point $A(V_1, P_1)$ is located on the P-V curve. Tangent at point A, cuts the V-axis at point B. AO is the line joining the point A to the origin O of PV diagram. BC is perpendicular to OB. Then



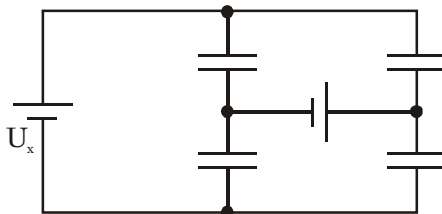
- (A) The coordinates of point C are $\left(2V_1, \frac{P_1}{2}\right)$
- (B) The coordinates of point C are $\left(\frac{3}{2}V_1, \frac{2}{3}P_1\right)$
- (C) Area of the triangle AOB is equal to nRT
- (D) Area of the triangle AOB is equal to $\frac{nRT}{2}$

Space for Rough Work

5. Given figure shows how the temperature of water and ice mixed with each other at different temperatures, changes with the amount of heat (Q) evolved by the water or absorbed by the ice. The scale corresponding to heat (Q) is arbitrary. Which of the following is/are conclusions drawn from the graph ?



- (A) The mass of water is double that of ice.
 (B) The equilibrium temperature will be 0°C with three-fourth of the ice melted.
 (C) The equilibrium temperature will be 0°C with one fourth of the water frozen.
 (D) The equilibrium temperature will be 10°C .
6. In the circuit in Figure, the "horizontal" battery has a voltage of 1 V, three of the four capacitors have the same capacity and the last twice as big. What can be the voltage of a second, "vertical" battery of voltage U_x , so that at least one capacitor in this circuit remained uncharged? Before battery connections were made capacitors are not charged.



- (A) 1V (B) 1.5 V (C) $\frac{2}{3}$ V (D) 2 V

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

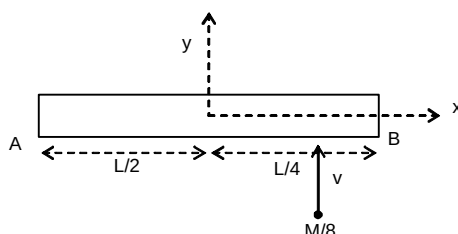
-									
●	●	●	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

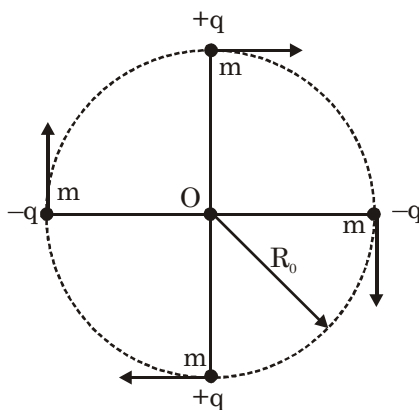
- There is an isolated planet having mass $2M$ and radius $2R$, where M and R are the mass and radius of the earth. A simple pendulum having mass m and length $2R$ is made to small oscillations on the planet. Find the time period of SHM of pendulum (in second).
(take $\pi = 3.00$, $g = 10 \text{ m/s}^2$, $R = 6400 \text{ km}$, $\sqrt{2} = 1.41$)
- A uniform rod of mass m and length $L = 2m$ is clamped at mid point, vibrates in the first overtone, with the frequency of 5000 Hz . Find the density of the rod (in kg/m^3).
(Y is the young modulus of the material of the rod $= 2 \times 10^{11} \text{ N/m}^2$).

Space for Rough Work

3. A slender rod of mass M and length $L = 1\text{ m}$ is lying on the frictionless surface. A particle of mass $\frac{M}{8}$ moving perpendicular to the length of the rod, with speed v and strikes the rod at a distance $\frac{L}{4}$ from the centre of mass of the rod. Find the y coordinate (in m) of the end B in time $t = \left(\frac{4\pi L}{v}\right)$ (Assume that after collision particle comes to rest.)



4. In free space on a circle with radius R_0 , four point masses m are located at the vertices of the inscribed square, two of them carry charge $+q$, and the other two charge $-q$. At the initial moment all these particles have same speed in clockwise direction as shown. It is known that during the motion the minimum distance from any of the point masses to the centre O of the initial circle is R_1 ($R_1 < R_0$). Consider that at any time, the charges are at the vertices of square centered at point O . The action of gravitational forces can be neglected. Determine the speed (in m/s) of any particle at a position having a distance R_1 from the centre of the circle. Take $m = 2\sqrt{2} - 1 \text{ gm}$, $q = 1\mu\text{C}$, $R_0 = 2R_1 = 1 \text{ m}$.



Space for Rough Work

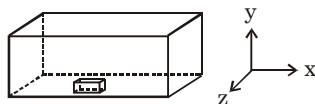
5. A point source of light emits a power of 100 W of monochromatic light with a wavelength of 248 nm. This source is at the focus of a convex lens of focal length 10 cm and aperture diameter 2 cm. On the other side of the lens we have a small metallic plate of radius 1 mm kept perpendicular to the optic axis such that its centre lies on the optic axis. The metallic plate has work function of 1.24 eV and a photoelectric efficiency of 0.1%. What is the initial photocurrent (in μA) from the plate after the source is switched on ?
6. If the beat frequency can be measured up to an accuracy of 0.1%, find the possible error (in cm/s) in measurement of speed of a vehicle moving with a approximate speed of 30 m/s towards a stationary source. The detector is located behind the source and is stationary. The source, detector and the moving vehicle are always on the same straight line. The speed of sound in air is known to be 330 m/s. The frequency is accurately known.

Space for Rough Work

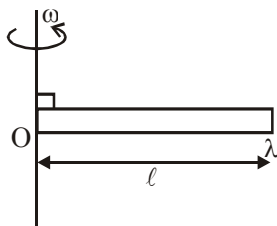
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\mathbf{v} = 5\hat{i} + 2t\hat{j}$ m/s. If the coefficient of friction between the surface of cube and block is 0.2. Then what is the force of friction between the block and cube (in N) ?



2. A uniformly charged rod of charge per unit length λ , is rotated about an axis as shown in the figure with a constant angular velocity $\omega = \frac{6}{(\lambda \ell^3)}$. Find the magnetic moment of the system.

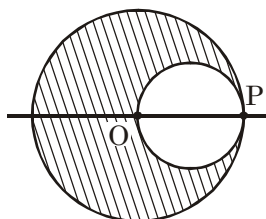


Space for Rough Work

3. A particle of mass 'm' having collided with a stationary particle of mass 'M' deviated by an angle $\frac{\pi}{2}$ rad where as the particle of mass 'M' moving at an angle of 30° to the direction of the initial motion of the particle 'm'. If after the collision there is 40% loss in kinetic energy of the two particle system, find $\frac{M}{m}$.
4. Assume that the refractive index of atmosphere of the earth is uniform and equal to 1.003. Assume that the atmosphere extends to height of 80 km above the surface of the earth. The radius of the earth is 6400 km. Due to the refraction, the sun sets after it has gone just below the horizon. Find the angular position of sun below the horizon (in degrees) when we see the sun setting. Round off to nearest integer.

Space for Rough Work

5. A body is executing simple harmonic motion. At a displacement x from mean position, its potential energy is $E_1 = 1\text{J}$ and at a displacement y from mean position, its potential energy is $E_2 = 4\text{J}$. The potential energy E at a displacement $(x + y)$ from mean position is _____J
6. A transparent sphere of radius R has a cavity of radius $\frac{R}{2}$ as shown in figure. Find the refractive index of the sphere if a narrow parallel beam of light falling on left surface focuses at point P. If this refractive index is $\mu = \frac{3 + \sqrt{X}}{2}$, find x .



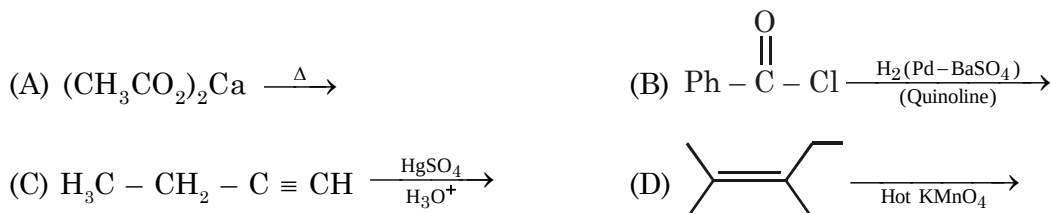
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
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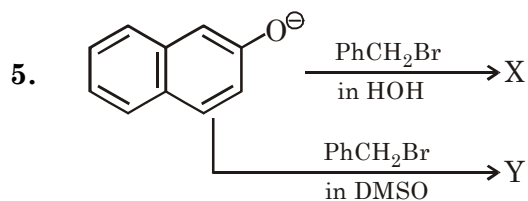
- Which of the following order is/are correct for hydrides of 15th group (EH_3) where $\text{E} = \text{N/P/As/Sb}$
 (A) Highest boiling point NH_3
 (B) Highest melting point SbH_3
 (C) Highest E-H distance SbH_3
 (D) Highest $\Delta H_{\text{disso. E-H}}$ NH_3
- Which of the following reaction gives aldehyde or ketone as final product?



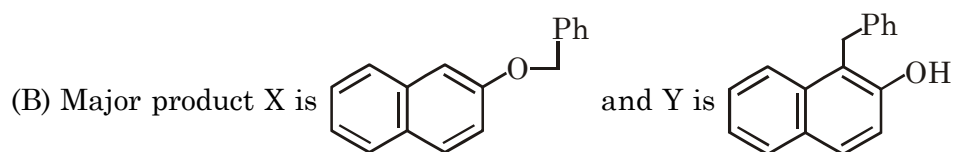
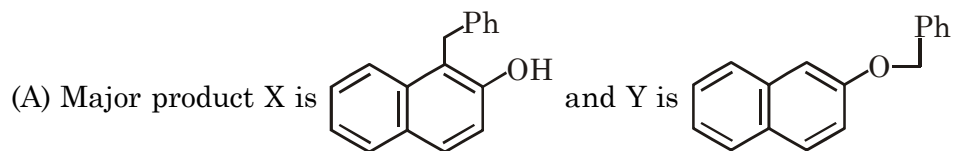
Space for Rough Work

3. 1 litre aq. solution containing sufficient sodium salt of 2,3 dimethyl butan-1,4dioicacid is electrolysed in an electrolytic cell using 9.65 mA current for 2×10^6 sec using inert Pt electrode with 100% current efficiency. Select the correct option(s) -
- (A) Mass of hydrocarbon formed at anode is 5.6 gm
 - (B) Mass of hydrocarbon formed at anode is 11.2 gm
 - (C) Gas evolved at cathode occupies 2.27 litre at STP
 - (D) Gas evolved at cathode occupies 1.135 litre at STP
4. Which of the following statement(s) is/are CORRECT?
- (A) D-penicillamine is used to remove excess of Cu through chelate therapy
 - (B) Desferrioxamine is used to remove excess of Fe through chelate therapy
 - (C) EDTA is used in the treatment of lead poisoning
 - (D) cis-platin and related compound effectively inhibit the growth of tumour

Space for Rough Work



correct statement(s) about above reactions and products is/are



(C) In HOH, O^- is highly solvated through hydrogen bond.

(D) In DMSO, no hydrogen bond is formed

6. Select the incorrect statement(s) :

(A) Stoichiometric defects always lowers density of solid.

(B) Defect only appears in ionic solids.

(C) Alkaline earth fluoride may show anion Frenkel defect.

(D) When AgCl is doped with Ag_2S , then anion vacancy created.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

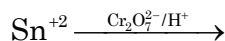
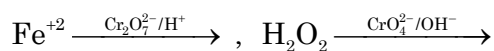
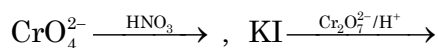
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●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- What is the sum of the square of the magnetic moments (spin only) in the following complex. $[\text{Ni}(\text{NH}_3)_2\text{Cl}_2]$, $[\text{RhF}_6]^{3-}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{PtCl}_4]^{2-}$, $[\text{Ag}(\text{CN})_2]^-$, $[\text{MnCl}_4]^{2-}$, $\text{Na}_2[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]$
- Identify total number of reagent sequence are used for following conversion :
 $\text{CH}_3 - \text{Cl} \longrightarrow \text{CH}_3 - \text{CH}_3$
 (a) Na / Dry ether (b) NaCN / H_3O^+ / NaOH / Electrolysis (c) LiAlH_4
 (d) MeMgCl / Dry ether (e) Mg-Dry ether / Me - I
 (f) Mg / Dry ether / CO_2 / LiAlH_4 / Red P + HI (g) Aq. NaOH
- When an electron jumps from n^{th} orbit to the 1^{st} orbit in H-atoms in one or more than one transitions, the the longest wavelength of the emitted photons is 1872 nm. The value of 'n' is (Given : $\frac{1}{R} = 91\text{nm}$)

Space for Rough Work

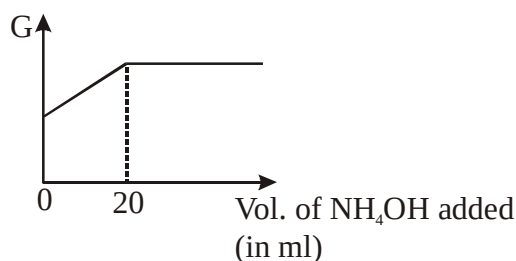
4. Find the number of reaction in which CrO_4^{2-} or $\text{Cr}_2\text{O}_7^{2-}$ does not act as oxidising agent.



5. How many of the following can be used to convert ethanol into ethanal only

- (a) $\text{H}^+\text{KMnO}_4/\Delta$ (b) Cu/Δ (c) NBS (d) $\text{DMSO} + \text{NaHCO}_3/\Delta$
 (e) PCC (f) $\text{H}^+\text{K}_2\text{Cr}_2\text{O}_7/\Delta$ (g) HIO_4/Δ (h) Schiff reagent

6. In 10 ml, 0.2M CH_3COOH ($K_a = 10^{-5}\text{M}$) aq. solution, 0.1M NH_4OH ($K_b = 10^{-5}\text{M}$) is gradually added and following conductance graph is observed.



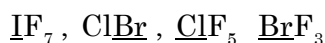
If pH of CH_3COOH solution initially is pH_1 & on adding 10 ml base it is pH_2 , calculate $(\text{pH}_2 - \text{pH}_1)$. [Given : $\log 2 = 0.3$]

Space for Rough Work

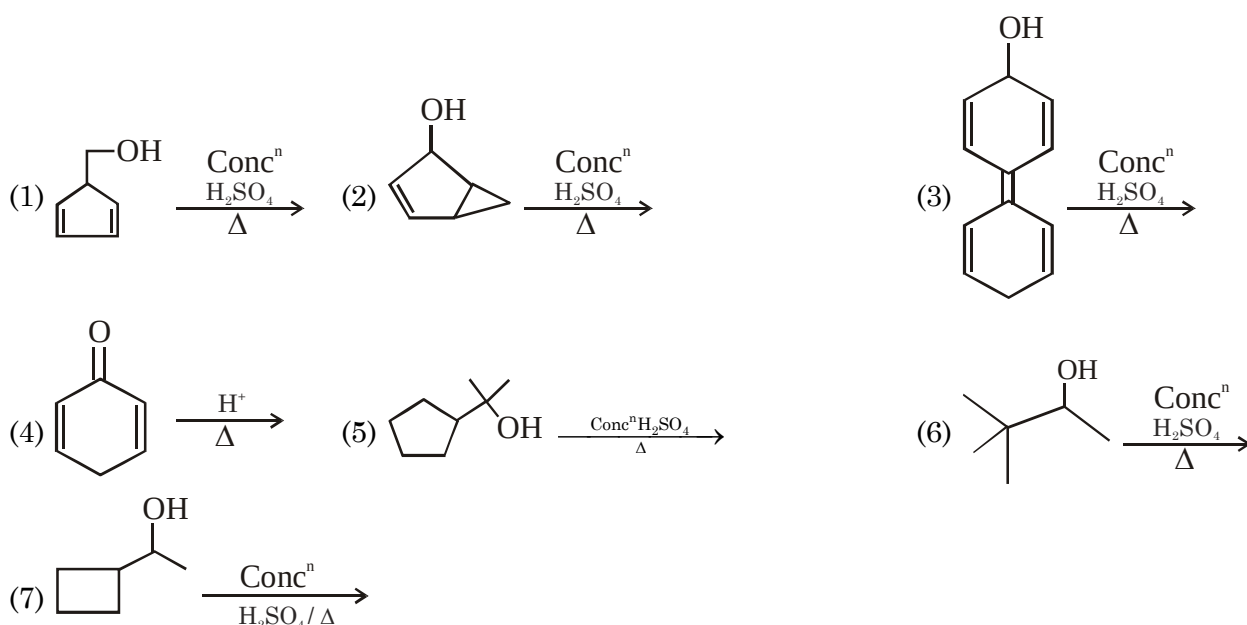
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Which of the following interhalogen on hydrolysis does not form oxyacid of their underliend atom.



2. Identify total number of reactions which give Aromatic product ?



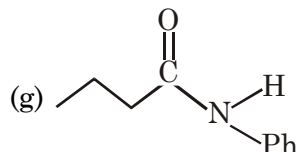
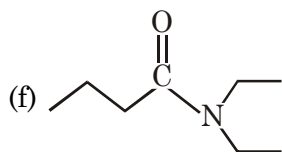
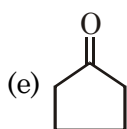
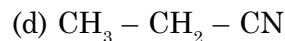
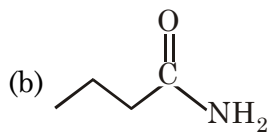
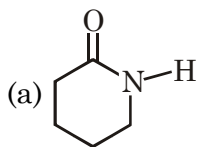
3. 'A' decomposes to give 'X' with a rate constant of $6.93 \times 10^{-2} \text{ min}^{-1}$. The reactant 'A' in the presence of catalyst gives 'Y' by a second order mechanism with a rate constant of $0.02 \text{ M}^{-1} \text{ min}^{-1}$. In order that half life of both the processess is same, one should start with an initial concentration of 'A' as 'C' M. The value of 'C' is.

Space for Rough Work

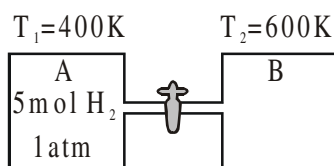
4. Find the number of oxides of halogens exist only at low temperature.

Br_2O , ClO_2 , Cl_2O , BrO_2 , BrO_3 , Cl_2O_7 , I_2O_7 , I_2O_5

5. Which of the following compounds form 2°-amines on reaction with LiAlH_4



6. Two containers A and B of equal volume maintained at 400 K and 600 K are connected with a narrow tube of negligible volume as shown in figure.



Find number of moles of H_2 gas in container A, when the stop cock is opened.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $\int \frac{\sec^2 x}{\sqrt{a \sec^2 x - b \tan^2 x}} dx$ is -

(A) $\frac{1}{\sqrt{b-a}} \sin^{-1} \left(\tan x \sqrt{\frac{b-a}{a}} \right) + c$ if $b > a > 0$

(B) $\frac{1}{\sqrt{b-a}} \log_2 \left(\tan x \sqrt{b-a} + \sqrt{a \sec^2 x - b \tan^2 x} \right) + c$ if $b > a > 0$

(C) $\frac{1}{\sqrt{a-b}} \sin^{-1} \left(\tan x \sqrt{\frac{a-b}{a}} \right) + c$ if $a > b > 0$

(D) $\frac{1}{\sqrt{a-b}} \log_e \left(\tan x \sqrt{a-b} + \sqrt{a \sec^2 x - b \tan^2 x} \right) + c$ if $a > b > 0$

(where c is integration constant)

Space for Rough Work

2. If there exists a circle 'C' whose one diameter is the join of foci of ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) and other diameter is the minor axis of E, then -
- (A) eccentricity of E is $\frac{1}{2}$
- (B) eccentricity of E is $\frac{1}{\sqrt{2}}$
- (C) If E is $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$ then director circle of circle 'C' is $x^2 + y^2 = 8$
- (D) If E is $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$ then director circle of circle 'C' is $x^2 + y^2 = 4$
3. Let p, q, r be distinct positive real numbers, such that exactly any two of the equations $px^2 + 2qx + r = 0$; $qx^2 + 2rx + p = 0$; $rx^2 + 2px + q = 0$ have a common root, say α , then which of the following statements is/are true -
- (A) α is real & negative
- (B) α is real & positive
- (C) remaining third quadratic equation has real roots
- (D) remaining third quadratic equation has imaginary roots

Space for Rough Work

4. The equation $a \cos 3x + b \cos 2x + c \cos x + \sin x = 0$ ($a, b, c \in \mathbb{R}$) has at least one root in $(0, 2\pi)$, for
- (A) $a = b = c = 0$
- (B) $a + b + c = 0$
- (C) a is positive and b & c must be of opposite sign
- (D) all values of a, b, c
5. Let 'O' be the origin and the line $y = (2 - \sqrt{3})x$ cuts the line $x = 1$ at point A. Right isosceles triangle OAB is constructed such that hypotenuse is OA and B lies in first quadrant. Which of the following statements is (are) true ?
- (A) length $AB = \sqrt{3} - 1$
- (B) length $OB = \sqrt{3} + 1$
- (C) $B \equiv \left(\frac{\sqrt{3}-1}{2}, \frac{3-\sqrt{3}}{2} \right)$
- (D) $B \equiv \left(\frac{\sqrt{3}+1}{2}, \frac{3+\sqrt{3}}{2} \right)$
6. If $A_1, A_2, \dots, A_n$ are n -distinct odd natural numbers not divisible by any prime greater than 5, then $\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \dots + \frac{1}{A_n}$ can never exceed -
- (A) 1
- (B) 2
- (C) 3
- (D) 4

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $ax + \int_0^x \ln t \, dt = \frac{x^2}{3}, (x \in \mathbb{R}^+)$ has exactly one solution, then the value of $(30(a + \ln 3))$ is equal to
2. The tangent at point P of a curve $y = f(x)$ meets the y-axis at A and the line parallel to y-axis through P meets the x-axis at B such that area of triangle OAB is 4 square units (O being the origin) and $f(1) = 6$ & $f(4) = 9$. If distance between point of maxima and minima on $y = f(x)$ is $\sqrt{d}$, then the value of 'd' is equal to

Space for Rough Work

3. Let $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in \mathbb{N}$ then find the value of $(a + b)$.
4. If P be point of intersection of line $\frac{x+1}{2} = \frac{y+2}{3} = \frac{z+3}{4}$ & plane $x + y + 4z = 6$ & A, B, C are point of intersection of given plane with X, Y and Z-axis respectively then find the distance of P from centroid of $\triangle ABC$.
5. The polynomial $P(x) = (1 + x + x^2 + \dots + x^{17})^2 - x^{17}$ has 34 complex roots of the form $z_k = r_k[\cos(2\pi a_k) + i \sin(2\pi a_k)]$, $k = 1, 2, 3, \dots, 34$ with $0 < a_1 \leq a_2 \leq a_3 \leq \dots \leq a_{34} < 1$ and $r_k > 0$. Given that $a_1 + a_2 + a_3 + a_4 + a_5 = \frac{p}{q}$, where p, q are relatively prime positive integers, then $(p + q)$ is equal to
6. A derivable function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the condition $f(x) - f(y) \geq \ell n \frac{x}{y} + x - y \forall x, y \in \mathbb{R}^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$.

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
 - The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
 - For each question, darken the bubble corresponding to the correct integer in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-

1. In a ΔABC (with usual notations), if $a \cos B - b \cos A = \frac{3}{5}c$, then $\frac{\tan A}{\tan B}$ is
 2. The value of $k(k \in \mathbb{R}^+)$ so that area of the bounded region enclosed between the parabolas $y = x - kx^2$ and $y = \frac{x^2}{k}$ is maximum, is
 3. In a 5 team tournament, each team plays one game with every other team. The probability of winning any game by any team is $\frac{1}{2}$ and there are no ties. Let $\frac{m}{n}$ be the probability that at the end of tournament there is no team with all wins or all defeats, m and n are relatively prime numbers. Then value $(2m - n)$ is
-

Space for Rough Work

4. If the normal at P(8,2) on the curve $xy = 16$ meets the curve again at Q. The angle subtended by PQ at origin is $\pi - \tan^{-1}\left(\frac{\lambda_1}{\lambda_2}\right)$ where λ_1 & λ_2 are relatively prime, then $\lambda_1 - 2\lambda_2 =$
5. Let A be the matrix of order 2×2 and commute with $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Number of matrices A, if its elements belong to the set $S = \{-1, 0, 1\}$, is
6. If $f(x) = \int \frac{3(\ln x)^2 + (\ln x)^4}{x(1 + (\ln x)^2 - (\ln x)^3)^2} dx$ and $f(1) = 0$, then the value of $|3f(e^2)|$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.
 17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

 Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

 Signature of the Invigilator